

Preparation of Papers for IEEE Sponsored Conferences & Symposia*

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Abstract—This electronic document is a *òliveÓ* template. The various components of your paper [title, text, heads, etc.] are already defined on the style sheet, as illustrated by the portions given in this document.

I. INTRODUCTION

Vision-Language-Action Models (VLAs) extends the vision-language understanding capabilities of Vision-Language Models (VLMs) to low-level robot control and contributes to the development of general-purpose robot policies. They take visual observations and language instructions as inputs and learn robot policies by modeling the distribution over robot actions. However, existing VLAs primarily focus on predicting robot actions and do not explicitly model how actions are carried out over time, limiting their ability to capture physical interactions in the real world. To address this limitation, World-Action Models (WAMs) further introduce video generation models to predict future dynamics. By leveraging the spatiotemporal and physical priors of video models, WAMs can better model interactions between the robot and the environment, providing stronger physical understanding for policy generation.

Existing WAMs can be broadly divided into two categories depending on whether future dynamics are explicitly constructed during inference. The first explicitly generates complete visual representations of future dynamics at inference, either by predicting future videos before action generation or by jointly modeling future videos and actions within a unified framework. The second directly predicts actions from observation representations encoded by video generation model. In this case, the learned interaction dynamics are encoded in the observation latent representation, rather than being explicitly realized as future videos. As shown in Fig. 1, such latent representations generalize poorly under OOD settings. Complete future prediction instead uses the learned dynamics to predict how the OOD scene will evolve, providing explicit future information for action generation and improving policy generalization. Yet, this explicit prediction requires dense pixel-level future generation, leading to substantial computational overhead and inference latency.

This overhead largely arises from generating complete pixel-level future videos, which capture both robot motion

and environment changes. However, action generation does not require all future information to be explicitly represented. Given the observation, future environment representations mainly reflect how the environment responds to robot motion, providing physical and interaction constraints on subsequent robot motion. For action prediction, the relevant information from these environment representations is ultimately reflected in the physical and interaction constraints on the robot's future motion. Therefore, instead of explicitly passing the complete future environment as the future representation to the action module, we can directly use the robot's future motion as a more compact representation, which not only reduces the computation and memory cost of inference, but also enables more candidate futures to be generated under the limited resource budget. More importantly, this compact representation provides each action candidate with a corresponding future motion, which can be used to evaluate and select among multiple candidates at inference time.

To achieve this, we adopt a Mixture-of-Transformer architecture with a Video Transformer for future video prediction and a Motion-Action Transformer for joint motion and action modeling. This design preserves the learning of future world dynamics in training while enabling Motion to serve as an explicit future representation. During training, the Video Transformer predicts future video tokens conditioned on the observation context without attending to Motion or Action. Motion and Action are jointly modeled and can attend to each other, while both are conditioned on the observation context and masked from the future video tokens. At inference time, future video generation is removed, while explicit future prediction is realized by Motion. The Video Transformer encodes the observation context in a single forward pass, and the Motion-Action Transformer then jointly predicts Motion and Action conditioned on the resulting representation. Since Motion is generated jointly with Action, each sampled action candidate is accompanied by a corresponding prediction of its execution process. We use this predicted Motion as the representation of each candidate for inference-time selection, and introduce a motion-based verifier to evaluate and rank multiple Motion-Action candidates. In this way, Motion provides a compact explicit future representation for both action generation and candidate selection, without requiring full future video generation at inference time. Our contributions are threefold:

- We introduce Motion as a compact explicit future representation for WAMs and design a Mixture-of-Transformer architecture that jointly learns future video prediction and Motion-Action generation. This allows

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the model to preserve explicit informative future prediction at inference time without generating dense future videos.

- Low computational cost of Motion enables efficient multi-candidate future prediction under a limited inference budget. Building on this property, we introduce a motion-based verifier for selecting the most suitable candidate.
- We conduct extensive experiments in simulation and real-world robotic tasks, demonstrating strong policy performance, substantially improved inference efficiency, and the effectiveness of Motion for multi-candidate generation and selection.

II. RELATED WORK

A. World Action Models

Recent advances in video foundation models have established World Action Models as an emerging paradigm for robotic manipulation, where future world dynamics are jointly modeled with robot actions. Early efforts such as UWM [?] unify video and action diffusion within a shared architecture, enabling policy learning together with forward and inverse dynamics modeling. Building upon large-scale video generative priors, subsequent works including Cosmos Policy [?], DreamZero [?], LingBot-VA [?], and MotuBrain [?] further demonstrate the effectiveness of joint video-action modeling for generalizable robot control. Beyond predicting RGB futures, recent studies have explored richer or more structured representations of world dynamics. X-WAM [?] extends WAMs to 4D world modeling by jointly predicting multi-view RGB-D observations and actions, while WSA₁ [?] explicitly incorporates future 3D scene dynamics into robot control. RepWAM [?] investigates representation-centric future modeling through semantic visual-action tokens, and OA-WAM [?] decomposes future states into object-addressable representations. Other works further strengthen the causal connection between actions and predicted futures, such as SelfWAM [?], which conditions future visual prediction on executed actions and introduces robot-centric future supervision. Collectively, these studies demonstrate a growing trend from conventional observation-to-action prediction toward explicitly modeling the future dynamics underlying embodied interaction.

B. Efficiency of World Action Models

Despite their strong performance, WAMs inherit the substantial computational cost of future video generation, motivating increasing attention toward efficient world action modeling. Existing approaches can be broadly categorized into three directions. The first replaces explicit visual reconstruction with more compact latent or representation-space future prediction. For example, LaWAM [?] predicts latent visual subgoals instead of reconstructing complete future observations, while JEPA-WAM [?] models future task-relevant representations through joint-embedding prediction. The second direction employs distillation to compress the iterative generation process into fewer inference steps. Methods such

as Flash-WAM [?] distill multi-step video-action generation into efficient few-step predictors, substantially reducing inference latency. The third direction explores asynchronous generation and computation, where action prediction does not need to wait for the entire visual denoising process. For example, X-WAM [?] decouples the generation schedules of visual and action streams, while other approaches employ asynchronous prediction and execution to further reduce control latency. Although these methods substantially improve WAM efficiency, they primarily focus on accelerating or compressing the original

III. METHOD

A. Problem Definition

Future representation. The role of world modeling is to predict how the world evolves in response to robot actions. Given observation latent $\mathbf{z}_o$, language instruction ℓ , and robot state $\mathbf{s}$, we define the conditioning context as $\mathbf{c} = (\mathbf{z}_o, \ell, \mathbf{s})$. Let $\mathbf{v}$ be the future visual latents and $\mathbf{a}_{1:H}$ an action chunk with horizon H . The basic formulation of World Action Models is:

$$p(\mathbf{a}_{1:H} | \mathbf{c}) = \int p(\mathbf{v} | \mathbf{c}) p(\mathbf{a}_{1:H} | \mathbf{v}, \mathbf{c}) d\mathbf{v}. \quad (1)$$

This modeling explicitly constructs future visual representations at inference time, introducing substantial overhead. To reduce this cost, Fast-WAM learns future dynamics through video prediction during training but removes future generation at inference time. Actions generation is conditioned on the observation representation $\mathbf{h}_{\text{obv}}^\theta$ produced by the trained Video Diffusion Transformer p_θ . Fast-WAM inference can be formulated as:

$$\mathbf{a}_{1:H} \sim p_\phi(\mathbf{a}_{1:H} | \mathbf{h}_{\text{obv}}^\theta, \ell, \mathbf{s}). \quad (2)$$

Although $\mathbf{h}_{\text{obv}}^\theta$ contains future dynamics learned from video prediction, its performance degrades under larger distribution shifts. In contrast, explicit future modeling can better adapt the learned dynamics to the current scene, but at a high inference cost. We therefore use future robot Motion as a compact explicit future representation, preserving future reasoning without generating complete future videos.

computational cost. We analyze the computational cost of future representations. For a prediction horizon T , a future video with L spatial tokens per step contains $N_v = TL$ tokens, whereas Motion with K structured motion elements per step contains only $N_m = TK$ tokens. In closed-loop robot control, both T and K are fixed. Under the Mixture-of-Transformer (MoT) paradigm used by WAMs, we consider a Transformer block with N self-attention tokens, N_ℓ language tokens, and hidden dimension d . Its dominant computational complexity is

$$C_{\text{block}}(N) = \mathcal{O}(Nd^2 + N^2d + NN_\ell d). \quad (3)$$

where the three terms correspond to linear projections and FFNs, self-attention, and cross-attention with language tokens as keys and values, respectively. Given N_0 existing

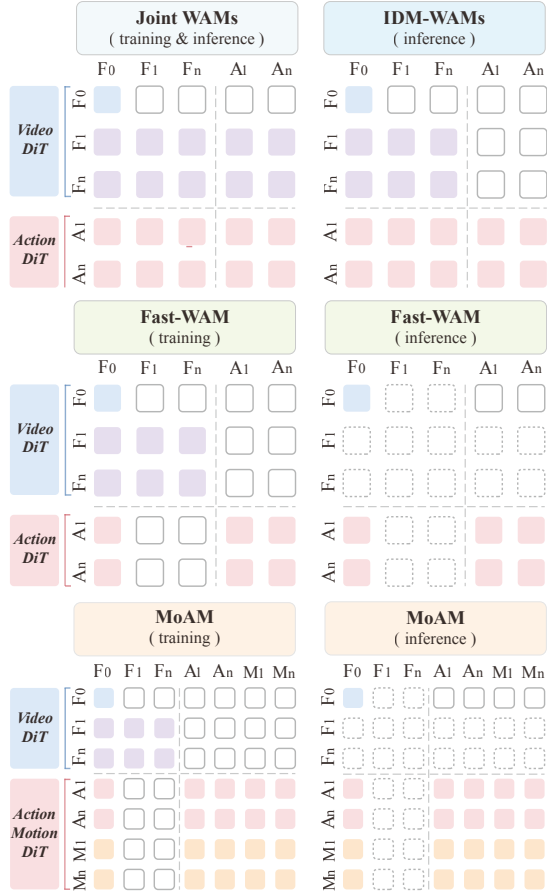


Fig. 1. To be done.

tokens and N_f additional future-representation tokens, the extra computational cost of explicit future modeling is

$$\Delta\mathcal{C}(N_f) = \mathcal{O}(N_f d^2 + (2N_0 N_f + N_f^2) d + N_f N_t d). \quad (4)$$

Substituting $N_f = N_v = TL$ for video and $N_f = N_m = TK$ for Motion gives

$$\begin{aligned} \Delta\mathcal{C}_{\text{video}} &= \mathcal{O}(L^2), \\ \Delta\mathcal{C}_{\text{motion}} &= \mathcal{O}(1), \end{aligned} \quad \text{with respect to } L. \quad (5)$$

Therefore, Motion replaces a dense visual future representation whose cost grows with the spatial granularity of the scene with a fixed-size structured motion representation, substantially reducing the inference cost of explicit future modeling.

1) AttentionMap:

B. MotionScaling

1) *Structured Motion Modeling*: Existing WAMs typically model actions jointly with future videos. However, during inference-time scaling, each candidate requires its own pixel-level future rollout, causing computational and memory costs to grow rapidly with the number of candidates. Therefore, we aim to design a compact representation that explicitly captures the future while reducing the computational cost of each candidate. Our key observation is that robot execution

is constrained by the robot's rigid mechanical structure. It can therefore be represented by the temporal evolution of the end-effector geometry, mainly determined by rigid transformations of the end effector and gripper articulation. Specifically, end-effector translation and rotation can be represented by the trajectories of points on the gripper, while gripper articulation can be represented by the distance between the fingers. Based on this observation, we represent each candidate future using structured end-effector motion instead of a high-dimensional pixel-level video. Specifically, we define three predefined keypoints on the end effector, located at the two fingertips and the gripper palm. We denote the positions of these three keypoints at future time step t under camera view v as

$$\mathbf{P}_t^v = [\mathbf{p}_{t,l}^v, \mathbf{p}_{t,r}^v, \mathbf{p}_{t,p}^v] \in \mathbb{R}^{3 \times 2}, \quad (6)$$

$\mathbf{p}_{t,l}^v$, $\mathbf{p}_{t,r}^v$, and $\mathbf{p}_{t,p}^v$ denote the projected positions under camera view v of the left fingertip, right fingertip, and palm keypoints, respectively. The trajectories formed by these keypoint positions over time

$$\mathbf{m} = \{\mathbf{P}_t^v\}_{t=1, v=1}^{T, V} \quad (7)$$

These keypoints define our structured motion representation. Their motion captures the overall movement of the end effector, while changes in their relative geometry capture gripper orientation and opening. This representation therefore captures both the global motion of the end effector and the relative motion of the fingertips.

However, treating the three keypoints as independent coordinates ignores their shared physical structure and may lead to inconsistent motion or implausible gripper geometry. We therefore constrain the temporal evolution of the keypoints using the mechanical structure of the end effector. For any set of keypoint positions ($\mathbf{P}_t^v$), we define

$$\mathbf{c}_t^v = \frac{\mathbf{p}_{t,l}^v + \mathbf{p}_{t,r}^v}{2}, \quad \mathbf{s}_t^v = \mathbf{p}_{t,r}^v - \mathbf{p}_{t,l}^v, \quad \mathbf{r}_t^v = \mathbf{p}_{t,p}^v - \mathbf{c}_t^v, \quad (8)$$

Here, $\mathbf{c}_t^v$ denotes the gripper center, $\mathbf{s}_t^v$ the fingertip span, and $\mathbf{r}_t^v$ the palm position relative to the gripper center. These three quantities characterize the overall end-effector structure.

Like actions, future motion is not uniquely determined by the current conditions, but follows a conditional distribution. We model the structured motion $\mathbf{m}$ using the same diffusion formulation as the action branch:

$$\mathbf{m} \sim p_\theta(\mathbf{m} \mid \mathbf{o}, \ell, \mathbf{s}). \quad (9)$$

The diffusion objective learns the conditional distribution of future motion, and the motion objective supervises its temporal changes. The structural constraint further regularizes the predicted keypoint geometry. The final objective for structured motion is

$$\mathcal{L}_{\text{mot}} = \mathcal{L}_{\text{diff}}^m + \lambda_{\text{str}} \mathcal{L}_{\text{str}} + \lambda_{\text{dyn}} \mathcal{L}_{\text{dyn}}. \quad (10)$$

2) MotionScaling Framework:

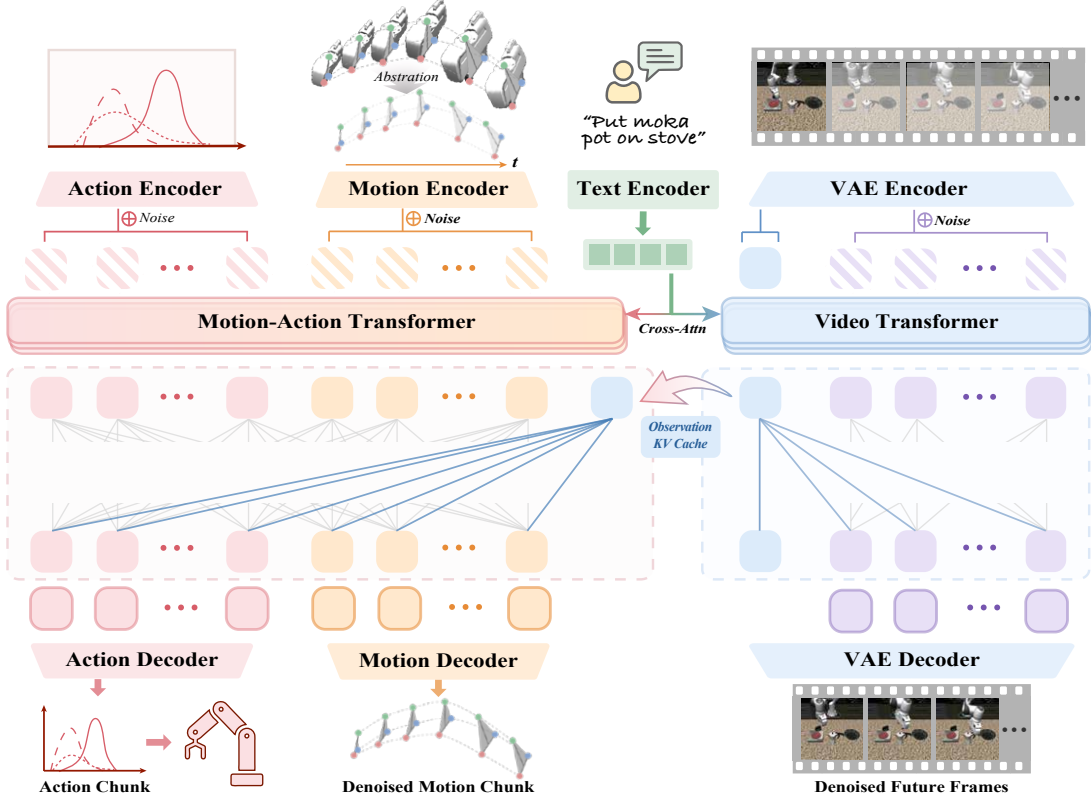


Fig. 2. To be done.

C. MoAM Architecture

MoAM adopts a dual-branch Transformer architecture, consisting of a Video Diffusion Transformer p_θ for future video modeling and an Action–Motion Diffusion Transformer p_ϕ for jointly modeling actions and robot motion. During training, both branches are jointly optimized given the current observation and language instruction. At inference time, the trained Video Diffusion Transformer p_θ only processes the observed frames to produce observation representations $\mathbf{h}_{\text{obv}}^\theta$, without generating complete future videos. The Action–Motion Diffusion Transformer p_ϕ then jointly generates actions and their corresponding future motions conditioned on these observation representations $\mathbf{h}_{\text{obv}}^\theta$. Given the task instruction and the current observation $\mathbf{O}$, the observed frames are first encoded into latent tokens $\mathbf{z}_{\text{obv}} = E_{\text{VAE}}(\mathbf{O})$. The future video frames, action chunk $\mathbf{a} = \mathbf{a}_1$, and structured motion $\mathbf{m}$ are then processed by their corresponding encoders and added with noise. As illustrated in the framework, for future video modeling, the Video Diffusion Transformer p_θ learns to denoise the noisy future video frames based on the current observation and task instruction, without receiving information from the Action–Motion branch. Actions and structured motion are jointly modeled in the Action–Motion Diffusion Transformer p_ϕ , where action and motion tokens access the observation representation $\mathbf{h}_{\text{obv}}^\theta$ through the corresponding keys and values in attention. We denote these observation keys and

values as

$$\text{KV}_{\text{obv}}^\theta = (\mathbf{K}_{\text{obv}}^\theta, \mathbf{V}_{\text{obv}}^\theta),$$

Then, the attention in the Motion–Action Transformer can be formulated as:

$$\mathbf{H}'_{am} = \text{Attn}(\mathbf{Q}_{am}, [\mathbf{K}_{\text{obv}}^\theta; \mathbf{K}_{am}], [\mathbf{V}_{\text{obv}}^\theta; \mathbf{V}_{am}]),$$

Here, $\mathbf{Q}_{am}$, $\mathbf{K}_{am}$, and $\mathbf{V}_{am}$ are obtained by projecting the action and motion tokens. In this way, action and motion exchange information within the same attention computation while directly accessing the current observation representation. This allows Action–Motion generation to leverage the dynamics learned by the Video Diffusion Transformer p_θ through future video prediction. The language instruction is incorporated separately through cross-attention in the two Transformer branches. At inference time, the current observation is passed through the Video Diffusion Transformer p_θ only once to obtain $\text{KV}_{\text{obv}}^\theta$, without further generating future videos. Action and Motion are initialized from random noise and jointly denoised in the Action–Motion Diffusion Transformer p_ϕ , conditioned on $\text{KV}_{\text{obv}}^\theta$, to recover the action chunk $\mathbf{a}_1$ and its corresponding future Motion. Finally, the generated action chunk is decoded by the Action Decoder into executable robot control commands.

1) *Motion-Based Verification*: To select a candidate Motion given the current observation and task instruction, we learn a conditional scoring function $s_\theta(I, \ell, \mathbf{m})$, where I , ℓ , and $\mathbf{m}$ denote the current image, language instruction, and a candidate future robot Motion, respectively. We construct ranking supervision from local Motion segments in successful demonstrations, using the preference that a demonstrated Motion should be ranked above its geometrically perturbed variants as a training prior.

Specifically, given a demonstrated Motion $\mathbf{m}^+$, we construct a negative sample $\mathbf{m}^-$ while keeping the initial robot state unchanged. The negative samples are generated by reducing the geometric variation of the Motion relative to the start-to-end reference motion, applying a smooth spatial offset to the subsequent Motion, or fixing a suffix of the Motion to simulate early stopping. These transformations introduce differences in motion shape, final spatial state, and motion completeness, respectively, providing relative preference supervision between positive and negative Motions under the same image and instruction.

At inference time, the candidate Motion is selected by

$$\arg \max_{\mathbf{m} \in \mathcal{M}} s_\theta(I, \ell, \mathbf{m}).$$

The score is used only for relative comparison among candidate Motions and is not interpreted as an absolute measure of task progress or success probability.

IV. EXPERIMENT

A. Experimental Setup

1) *Benchmark*: We evaluate MoAM and other baselines on the standard LIBERO benchmark, the out-of-distribution benchmark LIBERO-Plus, and real-world single-arm manipulation tasks, using task success rate as the primary evaluation metric.

LIBERO. LIBERO is a standard benchmark for language-conditioned robotic manipulation, comprising four task suites that cover spatial reasoning, object-centric manipulation, goal-conditioned tasks, and long-horizon tasks. Each suite contains 10 tasks and provides 500 demonstration trajectories. We perform 50 rollouts for each task and report the task success rate.

LIBERO-Plus. LIBERO-Plus extends LIBERO by introducing controlled perturbations to the original tasks, providing a systematic evaluation of policy robustness under out-of-distribution conditions. The benchmark includes seven types of distribution shifts involving camera viewpoints, robot initial states, language instructions, lighting conditions, background textures, sensor noise, and object layouts. For each perturbation type, we uniformly sample 100 evaluation instances from the four LIBERO task suites, resulting in a total of 700 OOD evaluation tasks. All methods are evaluated directly using policies trained on LIBERO, without any additional training or adaptation.

Real-World Evaluation. We conduct real-world experiments on a Franka Research 3 robot, using Intel RealSense D435 depth cameras mounted at the wrist and from a third-person view for observation. We consider three representative

manipulation tasks: Pick Bananas, Close Drawer, and Stack Bowls. These tasks require stable grasping, accurate spatial positioning, and controlled interaction with the environment, covering a diverse range of real-world manipulation behaviors and providing a comprehensive evaluation of policy performance. We collect 100 demonstrations for each task and conduct 20 evaluation trials per task under the standard setting.

2) *Implementation Details*: We use the pretrained Wan2.2-5B [36] as the backbone, including its video DiT, text encoder, and video VAE. Following Fast-WAM, the action expert adopts the same Transformer architecture as the video branch with a reduced hidden dimension of 1024, resulting in approximately 1B parameters for the action expert and 6B parameters for the full model. We set the action horizon to ($H=32$), and structured Motion uses the same horizon. Video frames are temporally downsampled by $4\times$, resulting in nine video frames per chunk. Images from multiple cameras are spatially concatenated before being encoded by the video VAE. Video, action, and Motion are trained with a flow-matching objective using a logit-normal distribution over (t) as the noise schedule. During inference, we use 10 denoising steps with the classifier-free guidance scale set to 1.0. We optimize the model using AdamW with a learning rate of 1×10^{-4} , weight decay of 0.01, and a cosine learning-rate schedule. Training is performed in mixed precision with gradient clipping at 1.0.

All experiments are trained on four NVIDIA A100 GPUs. For both LIBERO simulation and real-world experiments, data from different tasks are mixed for joint training, and the model is trained for 20K optimization steps. For comparisons with Fast-WAM, Joint-WAM, and IDM-WAM, we use identical training and inference configurations across all methods, with differences restricted to their model architectures.

B. Main Results

1) *LIBERO*: As shown in Table I, MoAM achieves the best overall performance on LIBERO and remains consistently strong across all four suites. Compared with representative WAM-based methods, MoAM clearly improves over Fast-WAM. While Fast-WAM learns future dynamics during training but keeps them implicit at inference, MoAM explicitly models compact future robot motion without generating dense future videos. This improvement suggests that explicit future representations provide useful information beyond dynamics implicitly encoded in observation features. MoAM also outperforms LaWAM, which reduces future-modeling cost by representing future dynamics in latent space. Rather than compressing the complete future, MoAM focuses on the execution-relevant future, explicitly modeling robot motion under current scene and interaction constraints, thereby providing a more direct connection between future modeling and action generation. Moreover, MoAM achieves competitive or better performance than prior WAM-based methods such as LingBot-VA and Motus, further demonstrating the effectiveness of Motion as a compact explicit future representation.

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE LIBERO
BENCHMARK.

Method	Robo. P.T.	Spa.	Obj.	Goa.	Lon.	Avg.
<i>VLA-based Methods</i>						
OpenVLA	✓	84.7	88.4	79.2	53.7	76.5
OpenVLA-OFT	✓	97.2	97.8	96	96	96.75
UniVLA	✓	95.4	94.8	94.8	90.8	93.95
π_0	✓	96.8	98.8	95.8	85.2	94.1
π_0 -FAST	✓	97.8	97.8	88.2	61	86.2
$\pi_{0.5}$	✓	<u>98.8</u>	98.2	<u>98</u>	92.4	96.9
<i>WAM-based Methods</i>						
Lingbot-VA	✓	98.5	<u>99.6</u>	97.2	98.5	98.5
Motus	✓	96.8	99.8	96.6	97.6	97.7
Fast-WAM	×	96.6	99.2	94.6	95.6	96.5
LaWAM	✓	99	96	97.2	93.4	96.4
IDM-WAM	×	98.6	99.4	98.4	96.4	98.2
Joint-WAM	×	99	99.2	99	<u>97.8</u>	<u>98.75</u>
Ours	×	<u>98.8</u>	99.8	99	<u>97.8</u>	98.9

2) *LIBERO-Plus*: As shown in Table ??, MoAM achieves the best overall performance on LIBERO-Plus, demonstrating stronger robustness under diverse distribution shifts. Compared with *Fast-WAM*, the improvement is substantial, showing that dynamics learned through video prediction but kept implicit at inference are not sufficient under challenging OOD conditions. Explicitly modeling future robot Motion allows the learned dynamics to be instantiated according to the current scene, providing additional guidance for action generation. The gains are particularly clear under object, initial-state, lighting, and sensor perturbations, where changes in the observation can directly affect the expected execution process. MoAM also outperforms *LaWAM* in overall performance. While LaWAM represents future dynamics in a compact latent space, MoAM explicitly focuses on the execution-relevant robot future, providing a more direct representation for policy generation. Notably, MoAM does not dominate every individual perturbation, with LaWAM remaining stronger under background and language variations, but its consistently strong performance across most settings leads to the highest average success rate. These results further demonstrate the effectiveness of Motion as a compact explicit future representation, particularly for improving policy robustness under distribution shifts.

3) *Real-World Evaluation.*: For the real-world evaluation, we compare MoAM with two representative WAM paradigms: Motus, which explicitly predicts future visual dynamics jointly with actions, and Fast-WAM, which learns future dynamics during training but does not explicitly generate the future at inference time. As shown in Table II, MoAM achieves the highest average success rate across the three real-world manipulation tasks. Compared with Fast-WAM, MoAM shows substantially stronger performance overall, indicating the benefit of explicitly modeling future execution

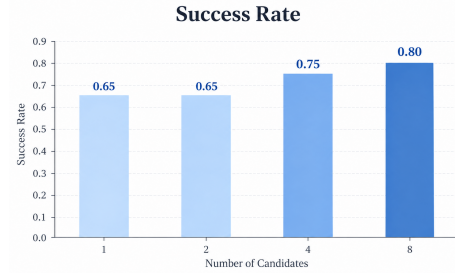


Fig. 3. Inference-time scaling with different numbers of Motion-Action candidates.

in real-world environments. Compared with Motus, MoAM achieves the same performance on Pick Banana, improves substantially on Stack Bowls, while performing slightly worse on Close Drawer. Overall, MoAM achieves a higher average success rate while replacing dense future video prediction with compact robot Motion, demonstrating that execution-relevant robot motion can serve as an effective explicit future representation for real-world policy generation.

4) *Inference-Time Scaling*: As shown in Fig. 3, MoAM exhibits a clear inference-time scaling trend as the number of Motion-Action candidates increases. Increasing the candidate set from 1 to 2 does not bring an immediate improvement, while larger candidate pools lead to consistent gains, reaching a success rate of 0.80 with eight candidates compared with 0.65 for a single candidate. This indicates that sampling multiple Motion-conditioned action hypotheses increases the opportunity to obtain a better execution plan, and the motion-based verifier can effectively exploit this additional candidate diversity. Importantly, because Motion is substantially more compact than dense future video, MoAM can scale the number of explicit future hypotheses at relatively low inference cost, enabling practical inference-time scaling for robot policy generation.

C. Ablation Study

- 1) *Motion Representation*:
- 2) *Verifier*:

V. CONCLUSIONS

APPENDIX

Appendixes should appear before the acknowledgment.

ACKNOWLEDGMENT

The preferred spelling of the word ‘acknowledgment’ in America is without an ‘e’ after the ‘g’. Avoid the stilted expression, ‘One of us (R. B. G.) thanks . . .’. Instead, try ‘R. B. G. thanks’. Put sponsor acknowledgments in the unnumbered footnote on the first page.

References are important to the reader; therefore, each citation must be complete and correct. If at all possible, references should be commonly available publications.

Method	Robo. P.T.	Objects	Camera	Initial	Light	Background	Sensor	Lang.	Avg.
OpenVLA-OFT	✓	73	65	27	<u>94</u>	89	70	87	72.14
UniVLA	✓	76	8	62	77	<u>90</u>	19	80	58.86
π_0	✓	81	<u>62</u>	40	92	81	<u>75</u>	70	71.57
π_0 -FAST	✓	73	<u>62</u>	26	72	76	70	78	65.29
Fast-WAM	×	82	42	<u>74</u>	85	63	72	75	70.43
LaWAM	✓	<u>84</u>	37	70	92	95	73	98	<u>78.43</u>
Ours	×	87	51	86	96	68	86	<u>96</u>	81.43

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS IN THE REAL WORLD
EVALUATION.

Method	Robo. P.T.	Pick Banana	Stack Bowls	Close Drawer	Avg.
Motus	✓	65	65	85	<u>71.67</u>
FastWAM	×	<u>30</u>	<u>85</u>	20	45
Ours	×	65	95	<u>80</u>	80

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